Foundation of Embodied AI: Course Map

Technical Tutorial

2026-10-20

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Revision	2026-10-20
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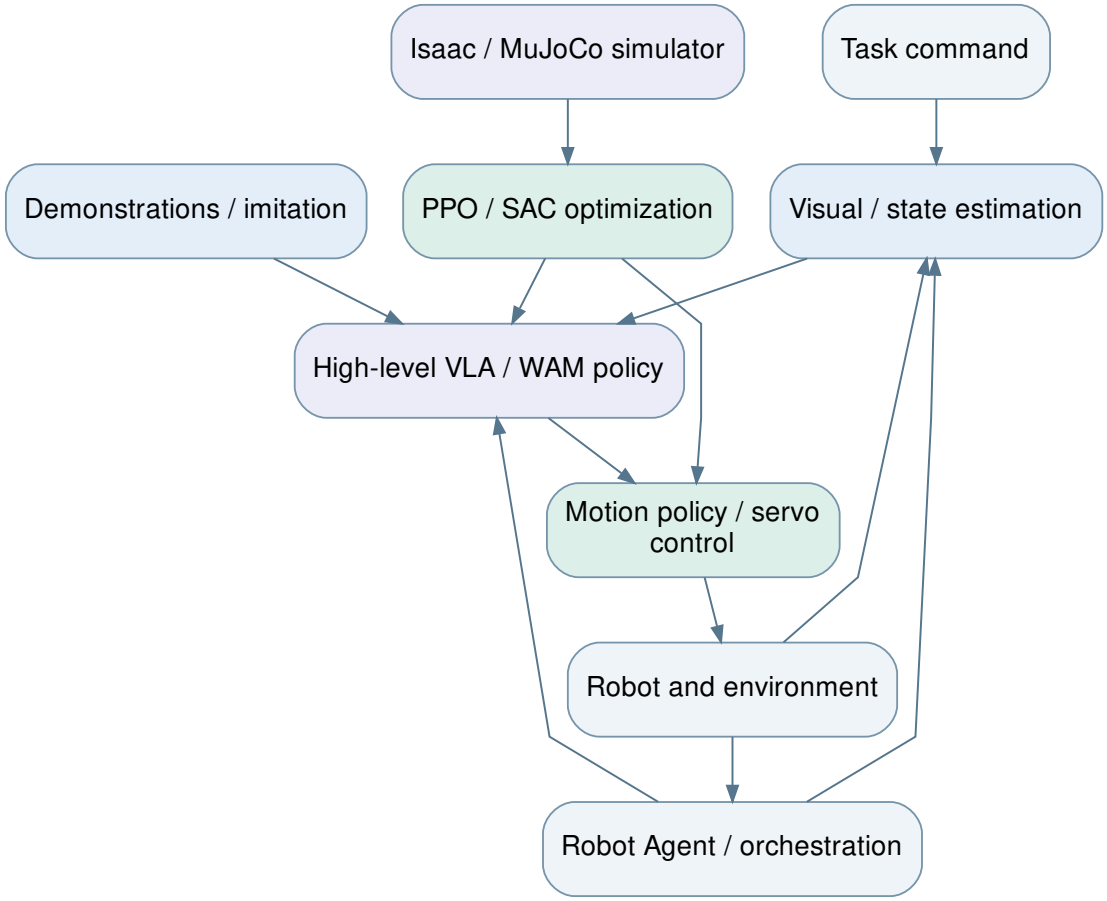
Six learning modules

Seventeen topics are organized as tools → calibration and deep learning → physics and perception → robot RL → demonstrations and generative policies → systems integration. Long-form derivations remain optional deep dives rather than strict prerequisites.

Module	Topics	Outcome
0. Orientation	Systems map and guide	Explain perception–policy–control loops
1. Tools	Git, ROS 2	Manage code and understand ROS communication and TF
2. Physics, perception and neural foundations	Calibration, Deep Learning Basics, vision foundation models, dynamics, Isaac / MuJoCo	Relate calibrated geometry, neural networks, physical models and simulation
3. Reinforcement learning	Motion RL, PPO, SAC	Design observations, actions, rewards and data collection
4. Demonstrations and models	Imitation Learning, Flow Matching, Efficient Training & Inference, RL Post-training, GR00T / SONIC, World Action Models	Learn generative and predictive policies and connect efficient inference with hierarchical controllers
5. Robot Agent integration	Robot Agent planning, tool use, verification and recovery	Orchestrate perception, policy and control for long-horizon tasks

Embodied AI system map



Reading paths

1. Beginners: orientation → essential Git and ROS 2 → calibration → deep learning → dynamics → vision foundation models → simulation.
2. Motion control: dynamics → simulation → deep learning → PPO / SAC → motion control RL.
3. VLA: calibration → deep learning → vision foundation models → Imitation Learning → Flow Matching → efficient training and inference → RL Post-training → GR00T / SONIC → World Action Models → Robot Agent.
4. Mathematical deep dives: full PPO, SAC, and Flow Matching derivations are reference material; hands-on learning does not require deriving everything first.

Tutorial overview

1. Git Basics

Git Basics — Learn Git's working tree, staging area and commit history, then follow a GitHub workflow covering branches, pull requests, conflict resolution and common recovery commands.

2. ROS 2 Basics

ROS 2 Basics — Understand nodes, topics, services, actions, QoS and TF. Connect sensors, state estimation, policy inference and controllers through ROS 2 interfaces.

3. Camera Calibration

Camera Calibration — Study camera intrinsics, distortion, extrinsics and hand–eye calibration. Relate RGB-D observations and coordinate transforms to robot-frame object poses.

4. Deep Learning Basics

Deep Learning Basics — Learn supervised objectives, gradient descent and backpropagation, then connect MLPs, Attention and Transformers to multimodal robot action heads. Understand pretraining, fine-tuning and the foundations required by downstream policy methods.

5. Vision Foundation Models

Vision Foundation Models — Compare segmentation, pose estimation, stereo depth and 3D foundation models. Follow four robotics workflows spanning known objects, stereo geometry, unknown shapes and human demonstrations.

6. Robot Dynamics

Robot Dynamics — Develop rigid-body motion and dynamics through coordinate frames, Jacobians, inertia and equations of motion. Connect contact, friction and control models to robot simulation.

7. Isaac / MuJoCo Simulation

Isaac / MuJoCo Simulation — Understand Isaac Sim, Isaac Lab and MuJoCo as simulation and training tools. Cover asset import, environment contracts, RL library selection, parallel throughput, deployment interfaces and a Tron2 + Revo3 scene case.

8. Motion Control RL

Motion Control RL — Study motion tracking, DeepMimic/AMP, universal controllers, teacher–student learning, RMA and residual RL. Connect these methods to legged and dexterous motion, recovery and Sim-to-Real.

9. PPO

PPO — Derive PPO-Clip and Actor–Critic learning from policy gradients, Bellman targets and GAE. Explain Gaussian policy updates, gradient isolation and asymmetric robot observations.

10. SAC

SAC — Trace Q-learning, TD/SARSA and DQN toward entropy-regularized RL. Derive soft Bellman equations and modern SAC with twin critics, reparameterization and automatic entropy tuning.

11. Imitation Learning

Imitation Learning — Introduce BC, DAgger and ACT, then connect Diffusion Policy and Flow Matching to action-chunk generation. Explain temporal ensembling and inference-time versus training-time RTC.

12. Flow Matching

Flow Matching — Derive conditional Flow Matching, velocity fields and the training objective. Explain how ODE integration transforms noise into conditional robot action sequences.

13. Efficient Training & Inference

Efficient Training & Inference — Explain Transformer compute and memory, FlashAttention, training optimizations and valid KV-cache reuse. Connect Flow Matching acceleration, action chunks and RTC to end-to-end VLA latency and continuous robot control.

14. RL Post-training

RL Post-training — Study RL refinement of pretrained Diffusion, Flow Matching and VLA policies. Compare trainable interfaces and Actor–Critic loops, then examine RLinf infrastructure, physical feedback and deployment boundaries.

15. GR00T & SONIC

GR00T & SONIC — Use GR00T and SONIC to explain System 1/System 0 humanoid control. Connect high-level action chunks, motion representations, whole-body policies and low-level execution.

16. World Action Models

World Action Models — Distinguish world models, VLAs and WAMs, then study DreamZero training and accelerated inference, Dream-1 cross-embodiment deployment, and VLA/WAM memory, compute and latency trade-offs. Connect these methods to contact-rich control, recovery and evaluation.

17. Robot Agents

Robot Agents — Distinguish foundation models, coding agents and agent harnesses. Connect RAX exploration, ASPIRE trace repair and RPent toolkits, memory, Flash Mode and the RLinf data flywheel to reusable robot workflows.